

CALQUE

Clone a disk — Windows included — to a larger disk

User guide ▪ version 0.1.0

Calque: French for a tracing, a faithful line-for-line copy of a drawing.

Three steps:

choose the two disks → **read the plan** → **Clone**

then, once the new disk is in place: **Extend C:**.

Windows (while running) ▪ Linux (from a USB stick)

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1 What Calque is for

Your system disk is too small and you bought a larger one. Calque copies **all** of the old disk to the new one — Windows' hidden partitions included — so that the computer starts from the new disk **exactly as before**, without reinstalling Windows or your software. The space gained is then given to C: in one click.

Your current disk is protected

- The source disk is **never modified**: Calque reads it and only writes to the target.
- The running system disk can never be chosen as the target.
- Before erasing the target, Calque shows what it will become and asks twice.
- Every copied block is **read back and compared**: a faulty USB link is detected.
- An interruption (cancel, unplugged cable, failure) leaves the target **empty**: it cannot be mistaken for a complete clone.

2 Before you start

1. **Back up your important documents** elsewhere (stick, external disk, cloud). Calque does not touch the source, but a backup remains the rule before any disk operation.
2. **BitLocker**: if C: is encrypted (common on recent laptops, "device encryption"), decrypt it first (**Settings** → **Privacy & security** → **Device encryption**, or `manage-bde -off C:` in an administrator terminal) and wait until it finishes. Calque detects it and refuses otherwise.
3. **Connect the new disk**: inside the PC, or in a USB enclosure.
4. **Size**: the target must be at least as large as the part of the source used by partitions (a 500 GB disk goes onto 500 GB or more; Calque works it out and tells you).

Warning

USB enclosures and 4096-byte sectors. Some enclosures (mostly for large SATA disks) present the disk to Windows with 4096-byte sectors instead of 512. A copy written through such an enclosure **does not boot** once the disk is installed in the PC: the partition table is no longer read at the right place. This is a common cause of "successful" clones that do not boot. Calque compares the two disks and **refuses** in that case, with an explanation (figure 1). Fix: another enclosure, the disk connected inside, or cloning on Linux (section 5).

3 Installing and starting Calque

Windows

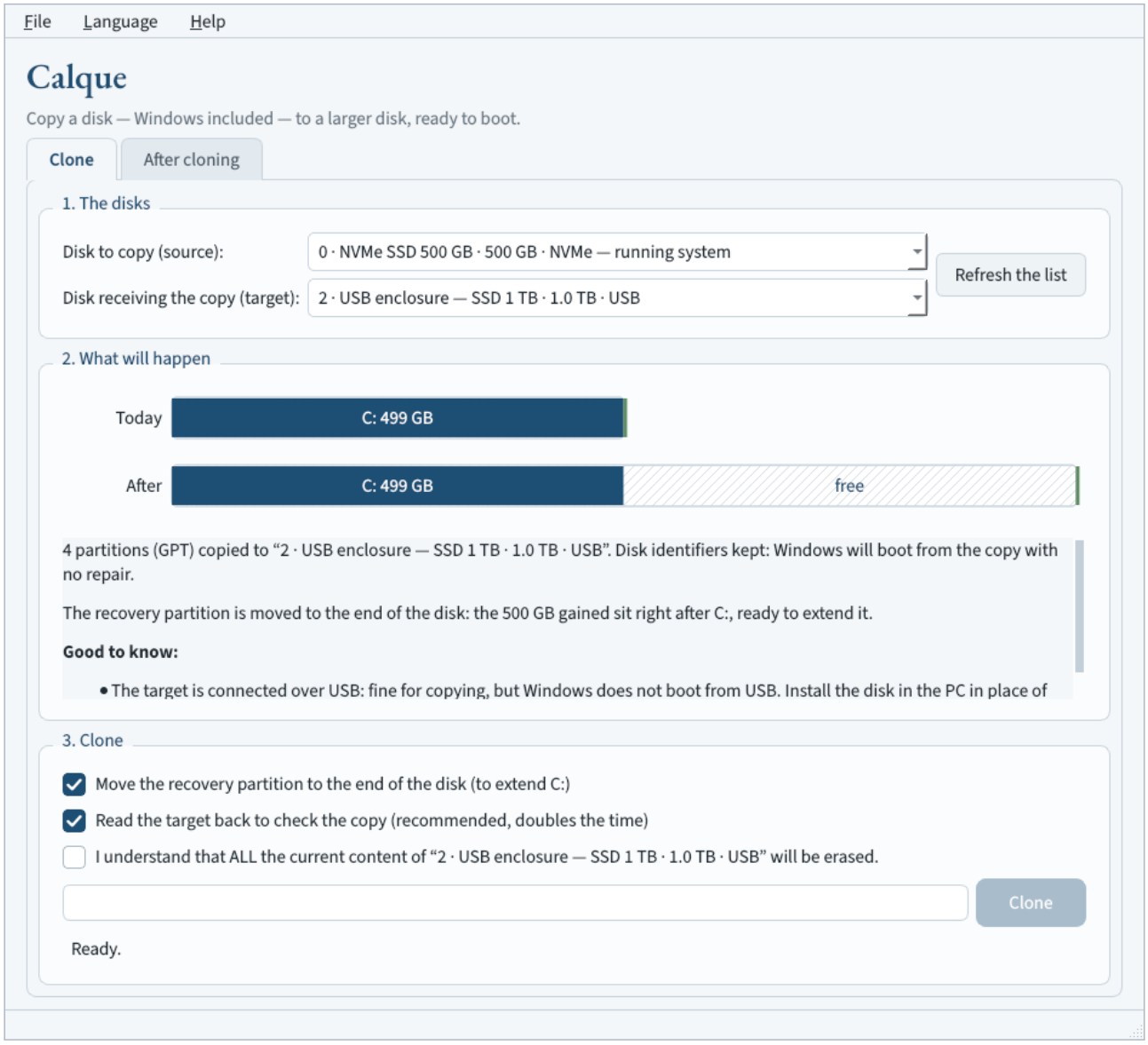
Download `Calque-Setup.exe` from the project's **Releases** page and run it (installed for your account, no special rights). Calque then asks, **every time it starts**, for administrator rights: reading and writing a whole disk requires them.

Linux

Download `Calque-linux-x86_64.tar.gz`, extract it, run `installer.sh`, then start Calque **with sudo**: `sudo ~/.local/bin/calque`. If the window does not open (a Wayland session refusing administrator windows), first type `xhost +si:localuser:root`, or use the command line (last section).

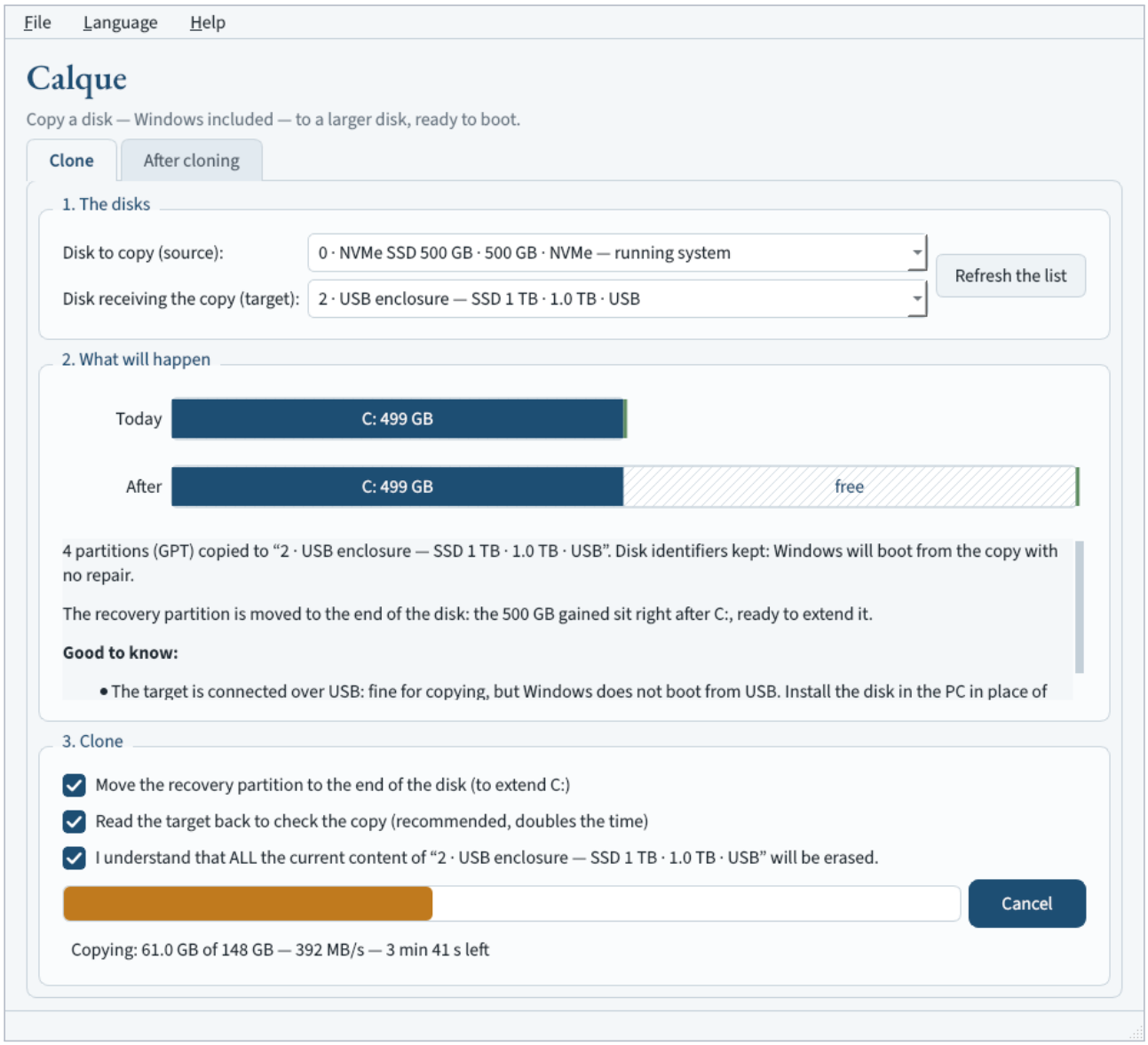
4 Cloning on Windows, without shutting down

Calque copies the disk **while Windows is running**. For a consistent copy, it first takes a **snapshot** (VSS) of each volume in use: a frozen image of C: at that moment. You can keep using the computer; what you change during the copy simply will not be on the clone. Easiest: close your documents and let it work.



1. **Source**: Calque picks the running system's disk by default.
2. **Target**: the new disk. The system disk never appears in this list. After plugging in a disk, click **Refresh the list**.
3. Read the **plan**: the "Today" bar shows the source, "After" the target, at the same scale. Windows' small partitions (EFI, reserved) are at the start; C: keeps its place; the **recovery** partition moves to the end of the disk, and the hatched "free" area is the space gained, right after C:.

4. Keep the **read-back** (check) ticked. It doubles the time but proves the copy.
5. Tick "I understand that ALL the current content… will be erased", click **Clone**, and confirm a second time.



Only the **used** blocks of C: are copied: a 500 GB disk holding 150 GB copies about 150 GB. Over USB 3, allow 5 to 15 minutes per 100 GB, and as much again for the check.

Good to know

Cancel stops cleanly: the target is left empty (no partition table) and the source disk has not changed. You can start again any time.

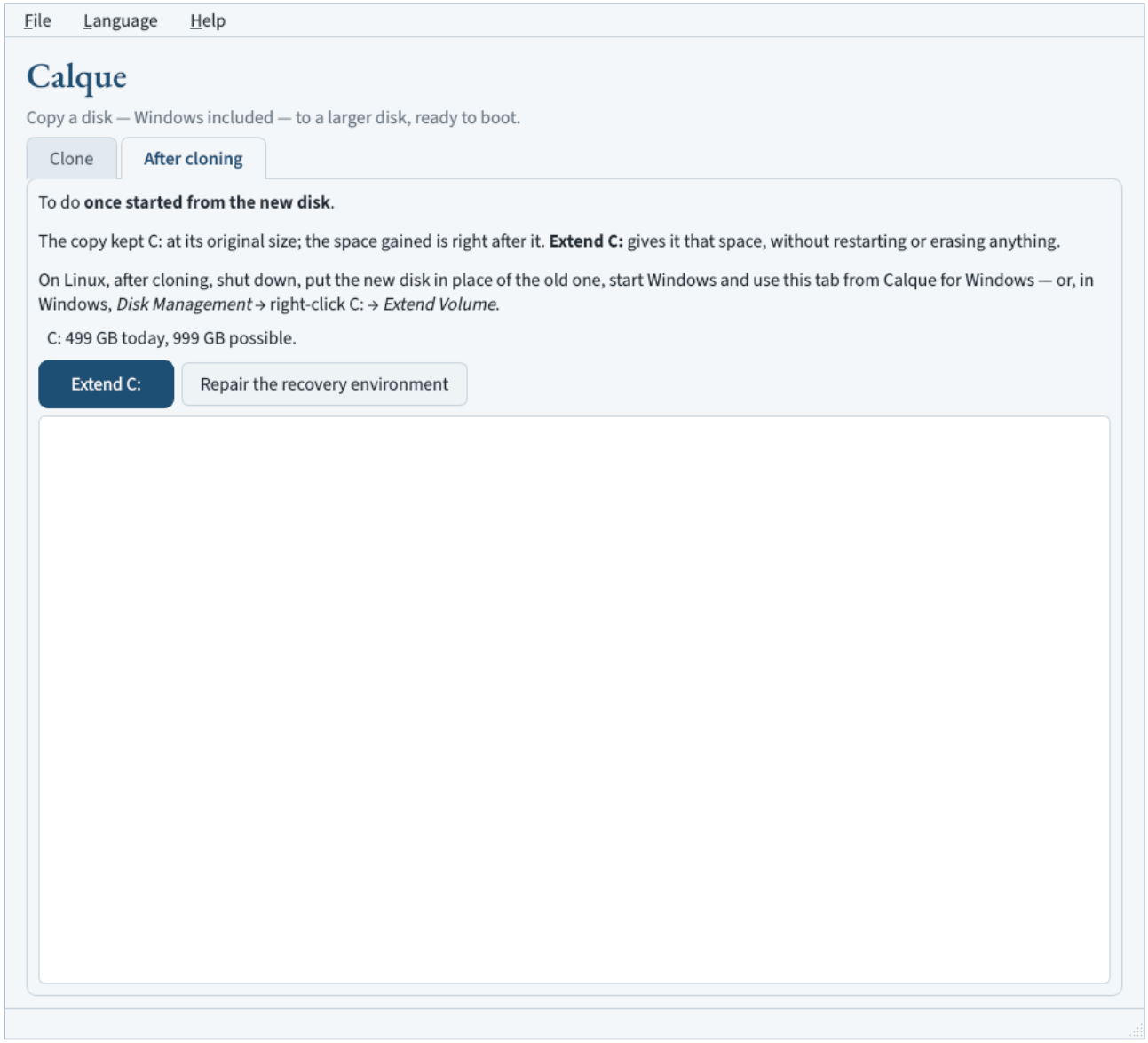
5 Cloning on Linux, from a USB stick

This is the safest method: Windows is **shut down**, its disk does not change during the copy, and no snapshot is needed. It is also the way out when the USB enclosure is the problem, since the new disk can be installed inside the PC next to the old one.

1. Prepare a "live" USB stick of a Linux distribution (Ubuntu, Linux Mint, Fedora…).
2. In Windows, turn off **fast startup** (**Control Panel → Power Options → Choose what the power buttons do**), then shut down: otherwise Windows leaves its disk in a "hibernated" state.
3. Start from the stick without installing Linux ("Try").
4. Install and start Calque with `sudo` (previous section), or on the command line: `sudo calque --list` then `sudo calque --clone /dev/nvme0n1 /dev/sdb`.
5. Choose source and target as on Windows. Calque refuses a source with a partition mounted read-write (on a live stick, none is by default).

6 After cloning

1. **Shut** the computer down.
2. **Put the new disk in place of the old one** (or unplug the old one). Windows does not start from a USB enclosure: the disk must be inside.
3. **Start.** Windows opens as before, from the new disk.
4. Run Calque, **After cloning** tab, and click **Extend C:**. C: takes all the free space, without restarting, in a few seconds.



Warning

Do not start with both disks connected the first time. The clone carries the same identifiers as the original (that is what lets it boot): Windows could start from one and mount partitions of the other. It actually sets the second one "offline" ("signature collision"). To reuse the old disk for data, connect it **afterwards** and erase it in **Disk Management**.

Recovery environment

The recovery partition was moved to the end of the disk. If `reagentc /info` (administrator terminal) shows the recovery environment as "Disabled", or a Windows update complains about it, the **Repair the recovery environment** button turns it off and on again, which registers it at its new place.

7 When it does not work

What you see	What to do
"512-byte sectors on the source, 4096 on the target"	The USB enclosure has the problem described on page 8: another enclosure, the disk inside, or Linux.
"The target is too small"	You need a target at least as large as stated.
"A volume is encrypted with BitLocker"	Decrypt C: first, wait until it finishes, try again.
"Dynamic disk or Storage Spaces"	Not supported (rare on a personal PC).
"Access denied"	Run Calque as administrator (Windows) or with `sudo` (Linux).
"Reading back found different blocks"	Faulty USB cable, port or enclosure, or a damaged target disk: change cable or port (preferably at the back of the PC) and start again.
The PC does not start from the new disk	In the BIOS/UEFI, choose the new disk ("Windows Boot Manager" entry) as the first boot device; make sure the old one is unplugged.

8 What Calque does, exactly

1. **Reads** the source's partition table (GPT or MBR), checked against its checksums.
2. **Plan**: the same layout on the target; partitions after the largest one (recovery) are moved to the end of the disk, aligned on 1 MiB.
3. **Snapshots** of the volumes in use (Windows), always deleted at the end.
4. **Erases** the start and the end of the target (old tables).
5. **Copies** partition by partition; for NTFS, only the blocks marked used in the volume's allocation map; each block gets a fingerprint.
6. **Reads back** the target and compares the fingerprints.
7. **Fixes** the boot sectors of moved volumes (partition position).
8. **Partition table** written last, with the **same identifiers** (disk and partition GUIDs in GPT, signature in MBR): the Windows boot manager refers to its partitions by them.

9 Command line

`calque --list`	lists the disks (number, model, size, sectors)
`calque --clone 0 2`	clones disk 0 to disk 2 (Windows)
`calque --clone /dev/nvme0n1 /dev/sdb`	the same on Linux
`--yes`	without asking for confirmation
`--no-verify`	without reading back
`--keep-layout`	without moving the recovery partition

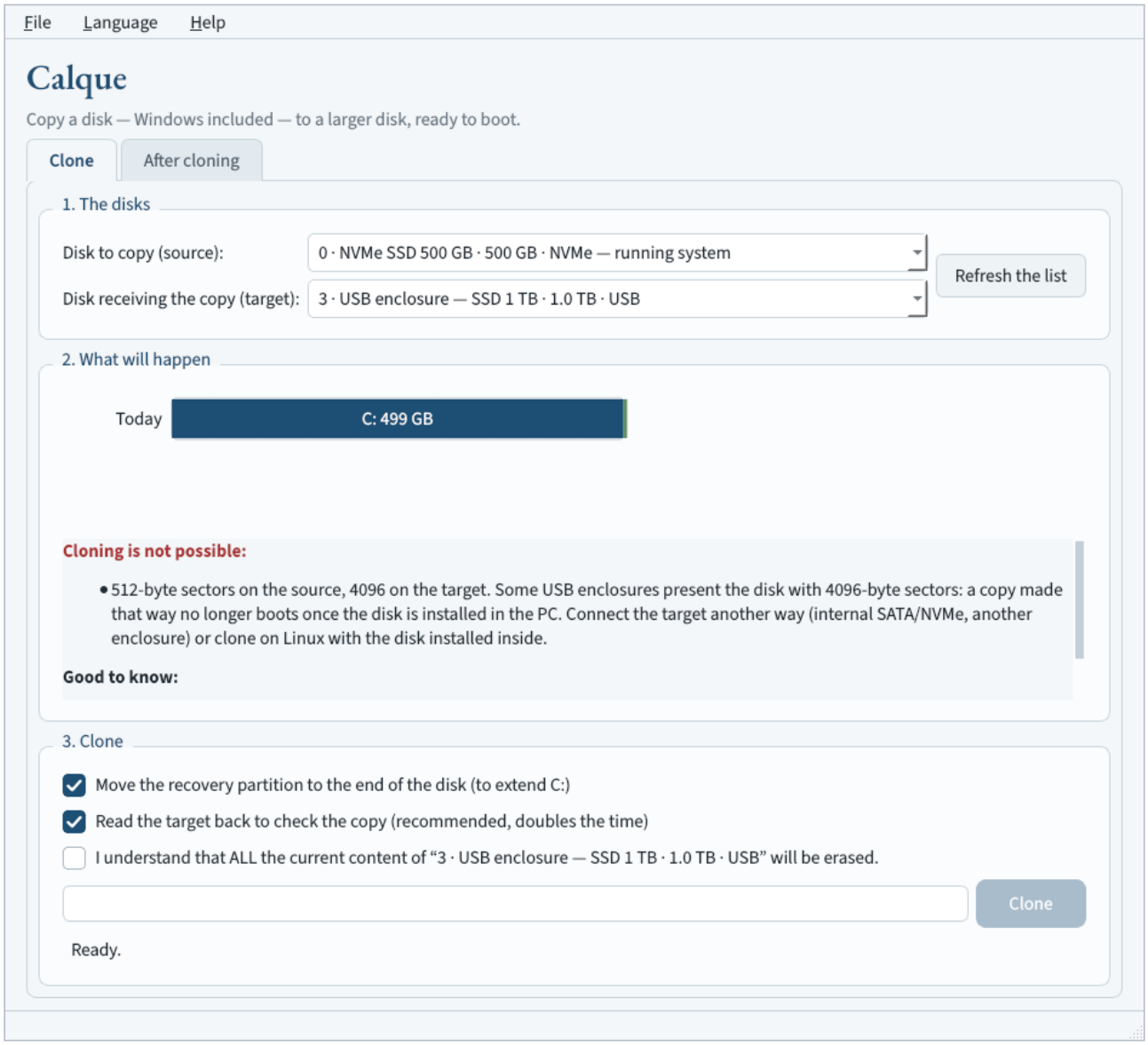


Figure 1: Target seen through a USB enclosure with 4096-byte sectors: cloning refused.